

The empirical recurrence for OEIS A297634: recovering a conjecture that is not written down

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Abstract

OEIS A297634 records “Empirical recurrence of order 68”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 100$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297634 is “Number of $n \times 5$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 0, 2 or 3 neighboring 1s.”. It has offset 1 and begins

13, 200, 1657, 17305, 201152, 2171757, 24145671, 273842684, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Jan 02 2018 11:57:22, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 100$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 100$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 200$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	20	c_{18}	5210529641	c_{35}	−727541405223	c_{52}	1117358284
c_2	−114	c_{19}	−3985676490	c_{36}	−647364298474	c_{53}	−22324899266
c_3	560	c_{20}	428016661	c_{37}	34291102968	c_{54}	−2325228196
c_4	−6648	c_{21}	−16141508942	c_{38}	257552751941	c_{55}	−881455406
c_5	29115	c_{22}	−6736105635	c_{39}	−145652133297	c_{56}	−841503970
c_6	−58695	c_{23}	−23304116855	c_{40}	−10011709450	c_{57}	887324908
c_7	558852	c_{24}	94577396464	c_{41}	464670535057	c_{58}	1066976032
c_8	−1918876	c_{25}	56903979684	c_{42}	217106939280	c_{59}	−56646192
c_9	−414293	c_{26}	141112561961	c_{43}	468939898836	c_{60}	213487444
c_{10}	−7344784	c_{27}	−144892202843	c_{44}	−100259329524	c_{61}	−1620464
c_{11}	33429564	c_{28}	−220309098028	c_{45}	−687468465554	c_{62}	−7381608
c_{12}	81546127	c_{29}	−416476457923	c_{46}	−319495066928	c_{63}	5975040
c_{13}	−189166167	c_{30}	10747756731	c_{47}	124316589366	c_{64}	−4965760
c_{14}	212994084	c_{31}	385563625984	c_{48}	172349960960	c_{65}	564864
c_{15}	−1397503622	c_{32}	778386481340	c_{49}	215446455637	c_{66}	−392320
c_{16}	2885769271	c_{33}	387513193358	c_{50}	−13327724237	c_{67}	17408
c_{17}	−4437674099	c_{34}	−309405986392	c_{51}	−53115086616	c_{68}	−15360

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68 + 4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

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